



Name: _____

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Learning Objectives

- Find GCD using the Euclidean Algorithm and LCM via the GCD
- Compute modular arithmetic and solve linear congruences
- Apply Fermat's Little Theorem to reduce large exponents
- Use prime factorization to count divisors

For the Euclidean Algorithm: always write each division step. For modular arithmetic: remainders must be in $[0, n-1]$.

1. Find $\gcd(48, 18)$ using the Euclidean Algorithm. Show all steps.

$$\gcd(48, 18) = ?$$

Answer: _____

2. Find $\text{lcm}(12, 18)$ using the formula $\text{lcm}(a,b) = ab / \gcd(a,b)$.

$$\text{lcm}(12, 18) = \frac{12 \cdot 18}{\gcd(12, 18)}$$

Answer: _____

3. Find the prime factorization of 360. Use it to find all divisors.

$$360 = ?$$

Answer: _____

4. Compute $17 \bmod 5$ and $-7 \bmod 3$. Explain what "mod" means.

$$17 \bmod 5 = ? \quad -7 \bmod 3 = ?$$

Answer: _____

5. Solve the linear congruence: $3x \equiv 7 \pmod{11}$.

$$3x \equiv 7 \pmod{11}$$

Answer: _____

6. Use the Euclidean algorithm to find $\gcd(252, 105)$ and express it as a linear combination (Bézout's Identity).

$$\gcd(252, 105) = 252s + 105t \text{ (Bezout)}$$

Answer: _____

7. Apply Fermat's Little Theorem to compute $3^{47} \bmod 23$.

$$3^{47} \bmod 23$$

Answer: _____



8. Determine if 97 is prime using trial division.

97 prime? $\sqrt{97} \approx 9.8$

Answer: _____

9. Find all solutions to $x^2 \equiv 1 \pmod{8}$.

$x^2 \equiv 1 \pmod{8}$

Answer: _____

10. Two bells ring together at time 0. Bell A rings every 12 minutes; Bell B rings every 18 minutes. When do they next ring together?

$\text{lcm}(12, 18) = ?$

Answer: _____





Problems 1–2: GCD then LCM — always in this order. Problem 5 (linear congruence): requires finding modular inverse.
Problem 7 (Fermat): reduce exponent modulo $p-1$ first.

Solutions

1. Find $\gcd(48, 18)$ using the Euclidean Algorithm. Show all steps.

$$\gcd(48, 18) = ?$$

$$\rightarrow 48 = 2 \cdot 18 + 12.$$

$$\rightarrow 18 = 1 \cdot 12 + 6.$$

$$\rightarrow 12 = 2 \cdot 6 + 0.$$

$$\rightarrow \text{Remainder is } 0 \rightarrow \gcd(48, 18) = 6.$$

Answer: $\gcd(48, 18) = 6$

2. Find $\text{lcm}(12, 18)$ using the formula $\text{lcm}(a, b) = ab / \gcd(a, b)$.

$$\text{lcm}(12, 18) = \frac{12 \cdot 18}{\gcd(12, 18)}$$

$$\rightarrow \gcd(12, 18): 18 = 1 \cdot 12 + 6, 12 = 2 \cdot 6 + 0 \rightarrow \gcd = 6.$$

$$\rightarrow \text{lcm}(12, 18) = 12 \cdot 18 / 6 = 216 / 6 = 36.$$

Answer: $\text{lcm}(12, 18) = 36$

3. Find the prime factorization of 360. Use it to find all divisors.

$$360 = ?$$

$$\rightarrow 360 \div 2 = 180. 180 \div 2 = 90. 90 \div 2 = 45.$$

$$\rightarrow 45 \div 3 = 15. 15 \div 3 = 5. 5 \text{ is prime.}$$

$$\rightarrow 360 = 2^3 \cdot 3^2 \cdot 5.$$

$$\rightarrow \text{Number of divisors} = (3+1)(2+1)(1+1) = 4 \cdot 3 \cdot 2 = 24 \text{ divisors.}$$

Answer: $360 = 2^3 \cdot 3^2 \cdot 5^1$

4. Compute $17 \bmod 5$ and $-7 \bmod 3$. Explain what "mod" means.

$$17 \bmod 5 = ? \quad -7 \bmod 3 = ?$$

$$\rightarrow 17 \bmod 5: 17 = 3 \cdot 5 + 2. \text{ Remainder} = 2.$$

$$\rightarrow -7 \bmod 3: -7 = (-3) \cdot 3 + 2. \text{ Remainder} = 2. (\text{remainder is always } \geq 0)$$

$$\rightarrow "a \bmod n" = \text{remainder when } a \text{ is divided by } n \ (0 \leq r < n).$$

Answer: $17 \bmod 5 = 2, \quad -7 \bmod 3 = 2$

5. Solve the linear congruence: $3x \equiv 7 \pmod{11}$.

$$3x \equiv 7 \pmod{11}$$

$$\rightarrow \text{Find } 3^{-1} \bmod 11 \text{ (the inverse of } 3 \bmod 11).$$

$$\rightarrow 3 \cdot 4 = 12 \equiv 1 \pmod{11} \rightarrow 3^{-1} = 4.$$

$$\rightarrow x \equiv 4 \cdot 7 = 28 \equiv 6 \pmod{11}.$$

Answer: $x \equiv 6 \pmod{11}$



6. Use the Euclidean algorithm to find $\gcd(252, 105)$ and express it as a linear combination (Bézout's Identity).

$$\gcd(252, 105) = 252s + 105t \text{ (Bezout)}$$

$$\rightarrow 252 = 2 \cdot 105 + 42.$$

$$\rightarrow 105 = 2 \cdot 42 + 21.$$

$$\rightarrow 42 = 2 \cdot 21 + 0. \gcd = 21.$$

$$\rightarrow \text{Back-substitute: } 21 = 105 - 2 \cdot 42 = 105 - 2(252 - 2 \cdot 105) = 5 \cdot 105 - 2 \cdot 252.$$

$$\rightarrow \text{Bézout: } 21 = 252(-2) + 105(5).$$

Answer: $\gcd(252, 105) = 21 = 5 \cdot 105 + (-2) \cdot 252$ (see steps)

7. Apply Fermat's Little Theorem to compute $3^{47} \pmod{23}$.

$$3^{47} \pmod{23}$$

$$\rightarrow \text{Fermat: if } p \text{ is prime and } \gcd(a,p)=1, \text{ then } a^{p-1} \equiv 1 \pmod{p}.$$

$$\rightarrow 23 \text{ is prime and } \gcd(3,23)=1 \rightarrow 3^{22} \equiv 1 \pmod{23}.$$

$$\rightarrow 47 = 2 \cdot 22 + 3. 3^{47} = (3^{22})^2 \cdot 3^3 \equiv 1 \cdot 27 \equiv 27 \pmod{23} = 4.$$

Answer: $3^{47} \equiv 3^3 \cdot (3^{22})^2 \equiv 27 \cdot 1 \equiv 4 \pmod{23}$

8. Determine if 97 is prime using trial division.

$$97 \text{ prime? } \sqrt{97} \approx 9.8$$

$$\rightarrow \text{Check divisibility by primes } \leq \sqrt{97} \approx 9.8: \text{ test } 2, 3, 5, 7.$$

$$\rightarrow 97 \text{ is odd (not } \div 2). 9+7=16 \text{ (not } \div 3). \text{ Ends in } 7 \text{ (not } \div 5). 97 \div 7 = 13.8... \text{ (not } \div 7).$$

$$\rightarrow \text{No prime } \leq 9 \text{ divides } 97 \rightarrow 97 \text{ is prime.}$$

Answer: 97 is prime

9. Find all solutions to $x^2 \equiv 1 \pmod{8}$.

$$x^2 \equiv 1 \pmod{8}$$

$$\rightarrow \text{Test } x = 0, 1, 2, 3, 4, 5, 6, 7 \pmod{8}:$$

$$\rightarrow 1^2=1 \equiv 1 \checkmark \quad 2^2=4 \quad 3^2=9 \equiv 1 \checkmark \quad 4^2=16 \equiv 0 \quad 5^2=25 \equiv 1 \checkmark \quad 6^2=36 \equiv 4 \quad 7^2=49 \equiv 1 \checkmark.$$

$$\rightarrow \text{Solutions: } x \equiv 1, 3, 5, 7 \pmod{8}.$$

Answer: $x \equiv 1, 3, 5, 7 \pmod{8}$

10. Two bells ring together at time 0. Bell A rings every 12 minutes; Bell B rings every 18 minutes. When do they next ring together?

$$\text{lcm}(12, 18) = ?$$

$$\rightarrow \text{They ring together again after } \text{lcm}(12, 18) \text{ minutes.}$$

$$\rightarrow \gcd(12, 18) = 6. \text{lcm} = 12 \cdot 18 / 6 = 36.$$

$$\rightarrow \text{They next ring together at } t = 36 \text{ minutes.}$$

Answer: $\text{lcm}(12, 18) = 36$ minutes

